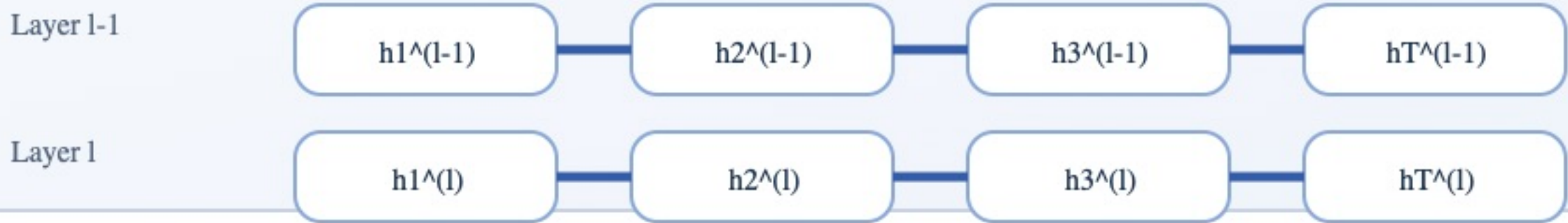


Future-Seed for prompt-time recurrent prefill

The baseline keeps each layer's default start state. Future-Seed only changes the start state of deeper layers; the within-layer scan and final decode stay causal.

Baseline recurrent stack

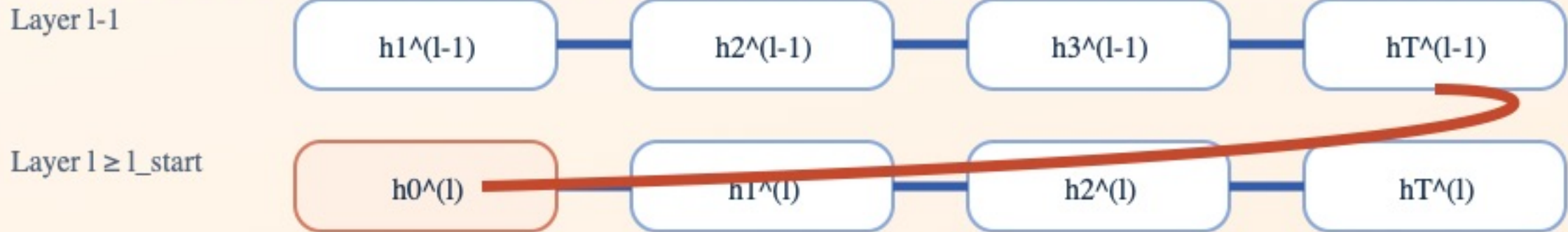
Each layer starts from its own default state and scans the prompt left to right.



No cross-layer feedback path.
Suffix evidence must survive only through the recurrent state itself.

Future-Seed recurrent stack

For layers $l \geq l_{\text{start}}$, the previous layer's terminal state seeds the next layer's start state through a small gate.



seed $h0^{(l)} \leftarrow g^{(l)} \odot hT^{(l-1)}$
Only this cross-layer initialization is new.

Decode path is unchanged: same left-to-right recurrent scan.